Chapter 8. Comparing Estimators

Criteria, sufficiency, UMVUE, asymptotic efficiency

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8.1 Criteria for comparing estimators

A good estimator of $\eta = \eta(\theta)$ gives values close to the truth. The most common measure of performance is the **mean squared error**

$$MSE(\hat{\eta}, \theta) = E_\theta\big[\{\hat{\eta}(X_1, \dots, X_n) - \eta(\theta)\}^2\big] = \operatorname{Var}_\theta \hat{\eta} + \{E_\theta \hat{\eta} - \eta(\theta)\}^2,$$

the variance plus the squared **bias**.

Example 8.1 (A clear winner)

For a $U(0, \theta)$ sample ($n \geq 2$), compare $\hat{\theta}^{\mathrm{MLE}} = X_{(n)}$ with $\hat{\theta}^{\mathrm{MME}} = 2\bar{X}$. Using $X_{(n)}/\theta \sim \mathrm{Beta}(n, 1)$,

$$MSE(X_{(n)}, \theta) = \frac{2\theta^2}{(n+1)(n+2)}, \qquad MSE(2\bar{X}, \theta) = 4 \operatorname{Var} \bar{X} = \frac{\theta^2}{3n}.$$

The MLE is better for every θ and every n, and its MSE decays like n^{-2} rather than n^{-1}.

Example 8.2 (No uniform winner)

For $N(\theta, 1)$, compare $\bar{X}$ (MSE $1/n$) with the silly estimator $\hat{\theta}^0 \equiv 0$ (MSE θ^2). If $|\theta| < 1/\sqrt{n}$, the constant has smaller MSE. Neither dominates the other. A more interesting competitor is the "pretest" estimator

$$\hat{\theta}^H = \begin{cases} 0, & |\bar{X}| < 1.96/\sqrt{n}, \\ \bar{X}, & |\bar{X}| \geq 1.96/\sqrt{n}. \end{cases}$$

Its MSE, $n^{-1}E\big[(Z + \sqrt{n}\theta)^2 \mathbf{1}\{|Z + \sqrt{n}\theta| \geq 1.96\}\big] + \theta^2 P(|Z + \sqrt{n}\theta| < 1.96)$ with $Z \sim N(0, 1)$, depends on θ only through $\sqrt{n}\theta$. Times n, it is below 1 near $\theta = 0$ but rises to about 2.5 when $\sqrt{n}|\theta| \approx 2$. For fixed $\theta \neq 0$, $n \cdot MSE \to 1$.

Since MSEs typically cross, we need criteria that do not depend on the unknown θ:

- **Maximum MSE**: compare $\max_\theta MSE(\hat{\eta}, \theta)$; an estimator minimizing it is **minimax**.
- **Bayes MSE**: for a weight (prior) function $\pi \geq 0$ on Ω, compare $r(\pi, \hat{\eta}) = \int_\Omega MSE(\hat{\eta}, \theta)\pi(\theta)\, d\theta$; a minimizer is a **Bayes estimator** for π.

For a positive target one sometimes uses the relative MSE $E_\theta(\hat{\eta}/\eta - 1)^2$ instead.

Example 8.3

For $N(\mu, \sigma^2)$, consider $\hat{\sigma}^2_c = c\sum(X_i - \bar{X})^2$, $c > 0$. Since $V = \sum(X_i - \bar{X})^2/\sigma^2 \sim \chi^2(n-1)$,

$$E_\theta(\hat{\sigma}^2_c/\sigma^2 - 1)^2 = E(cV - 1)^2 = c^2 \cdot 2(n-1) + \{c(n-1) - 1\}^2,$$

which does not depend on θ. Minimizing in c gives $c = 1/(n+1)$: the minimax relative-MSE estimator is $\frac{1}{n+1}\sum(X_i - \bar{X})^2$, neither the MLE ($c = 1/n$) nor S^2 ($c = 1/(n-1)$).

Another approach restricts the class of estimators. An estimator is **unbiased** if $E_\theta \hat{\eta} = \eta(\theta)$ for all $\theta \in \Omega$; then $MSE = \operatorname{Var}_\theta \hat{\eta}$.

Definition (UMVUE)

$\hat\eta^*$ is a **uniformly minimum variance unbiased estimator** (UMVUE) of $\eta(\theta)$ if (i) it is unbiased, and (ii) $\operatorname{Var}_\theta \hat\eta^* \le \operatorname{Var}_\theta \hat\eta$ for all θ and every unbiased $\hat\eta$.

Theorem 8.1 (Uniqueness)

If the UMVUE of $\eta(\theta)$ has finite variance, it is unique: if $\hat\eta_1, \hat\eta_2$ are both UMVUEs, then $P_\theta(\hat\eta_1 = \hat\eta_2) = 1$ for all θ.

Proof. $\hat\eta_3 = (\hat\eta_1 + \hat\eta_2)/2$ is unbiased, so $\operatorname{Var}\hat\eta_3 \ge \operatorname{Var}\hat\eta_1 = \operatorname{Var}\hat\eta_2$, i.e.

$$\tfrac14\{\operatorname{Var}\hat\eta_1 + 2\operatorname{Cov}(\hat\eta_1,\hat\eta_2) + \operatorname{Var}\hat\eta_2\} \ge \operatorname{Var}\hat\eta_2,$$

so $\operatorname{Cov}(\hat\eta_1,\hat\eta_2) \ge \operatorname{Var}\hat\eta_2$. Then $\operatorname{Var}(\hat\eta_1 - \hat\eta_2) = \operatorname{Var}\hat\eta_1 + \operatorname{Var}\hat\eta_2 - 2\operatorname{Cov}(\hat\eta_1,\hat\eta_2) \le 0$. A variable with zero variance and zero mean is 0 with probability one (Theorem 1.11 in Volume I). ■

8.2 Sufficient statistics

Can a part of the data, a statistic, carry all the information about θ?

Example 8.4

Two Bernoulli(θ) trials X_1, X_2, summarized by $Y = X_1 + X_2$. If $Y = 0$ or 2, the data are determined. If $Y = 1$,

$$P_\theta(X_1 = 0, X_2 = 1 \mid Y = 1) = \frac{(1-\theta)\theta}{2\theta(1-\theta)} = \frac12 = P_\theta(X_1 = 1, X_2 = 0 \mid Y = 1).$$

The original data can be "recreated" from Y by a coin flip, without knowing θ. So Y is as informative as (X_1, X_2).

Definition (Sufficiency)

A statistic $Y = u(X_1, \dots, X_n)$ is **sufficient** for $\theta \in \Omega$ if the conditional distribution of $(X_1, \dots, X_n)$ given $Y = y$ does not depend on θ, i.e. $P_{\theta_1}(\mathbf{X} \in A \mid Y = y) = P_{\theta_2}(\mathbf{X} \in A \mid Y = y)$ for all A, y, θ_1, θ_2.

Example 8.5

For n Bernoulli(θ) trials and $Y = \sum X_i \sim B(n,\theta)$,

$$P_\theta(X_1 = x_1, \dots, X_n = x_n \mid Y = y) = \frac{\theta^y(1-\theta)^{n-y}}{\binom{n}{y}\theta^y(1-\theta)^{n-y}} = \binom{n}{y}^{-1} \quad (\textstyle\sum x_i = y),$$

free of θ. So $\sum X_i$ is sufficient: all arrangements with y successes are equally likely.

Theorem 8.2 (Factorization)

$Y = u(\mathbf{X})$ is sufficient for $\theta \in \Omega$ if and only if there are functions k_1, k_2 with

$$\prod_{i=1}^n f(x_i;\theta) = k_1(u(\mathbf{x}),\theta)\, k_2(\mathbf{x}) \qquad \text{for all } \mathbf{x} \text{ and } \theta \in \Omega.$$

Proof. We prove the discrete case. ($\Leftarrow$) For $\mathbf{x}$ with $u(\mathbf{x}) = y$,

$$P_\theta(\mathbf{X}=\mathbf{x} \mid Y=y) = \frac{P_\theta(\mathbf{X}=\mathbf{x})}{\sum_{\mathbf{z}:u(\mathbf{z})=y} P_\theta(\mathbf{X}=\mathbf{z})} = \frac{k_1(y,\theta)k_2(\mathbf{x})}{k_1(y,\theta)\sum_{\mathbf{z}:u(\mathbf{z})=y} k_2(\mathbf{z})} = \frac{k_2(\mathbf{x})}{\sum_{u(\mathbf{z})=y} k_2(\mathbf{z})},$$

which does not depend on θ; for $u(\mathbf{x}) \neq y$ it is 0. ($\Rightarrow$) If Y is sufficient, $P_\theta(\mathbf{X}=\mathbf{x} \mid Y = u(\mathbf{x})) =: k_2(\mathbf{x})$ does not depend on θ, and with $k_1(y,\theta) = P_\theta(Y=y)$,

$$P_\theta(\mathbf{X}=\mathbf{x}) = P_\theta(\mathbf{X}=\mathbf{x}, Y=u(\mathbf{x})) = k_2(\mathbf{x})\,k_1(u(\mathbf{x}),\theta).$$

■

Example 8.6

(a) Poisson(θ): $\prod f = e^{-n\theta}\theta^{\sum x_i}\cdot\prod(1/x_i!)$, so $\sum X_i$ is sufficient.

(b) Exp(θ): $\prod f = \theta^{-n}e^{-\sum x_i/\theta}\cdot\prod \mathbf{1}\{x_i>0\}$, so $\sum X_i$ is sufficient.

(c) Gamma(α,β): $\prod f = \{\Gamma(\alpha)\beta^\alpha\}^{-n}(\prod x_i)^{\alpha-1}e^{-\sum x_i/\beta}\cdot\prod \mathbf{1}\{x_i>0\}$, so $(\sum X_i, \prod X_i)$ is sufficient for (α,β).

Theorem 8.3 (Exponential families)

If $f(x;\theta) = \exp\{\sum_{j=1}^k g_j(\theta)T_j(x) - A(g(\theta)) + S(x)\}$ for $x \in \mathcal{X}$, with support $\mathcal{X}$ not depending on θ, then $\left(\sum_i T_1(X_i), \ldots, \sum_i T_k(X_i)\right)$ is sufficient for θ.

Proof. $\prod f(x_i;\theta) = \exp\{\sum_j g_j(\theta)\sum_i T_j(x_i) - nA(g(\theta))\}\cdot\exp\{\sum S(x_i)\}\prod \mathbf{1}_{\mathcal{X}}(x_i)$; apply the factorization theorem. ■

Theorem 8.4

If Y is sufficient for θ and $W = g(Y)$ with g one-to-one, then W is sufficient.

Proof. $\{W=w\} = \{Y = g^{-1}(w)\}$, so $P_\theta(\mathbf{X}\in A \mid W=w) = P_\theta(\mathbf{X}\in A \mid Y = g^{-1}(w))$, which does not depend on θ. ■

Example 8.7

(a) $N(\mu,\sigma^2)$ is a two-parameter exponential family with $T=(x,x^2)$, so $(\sum X_i, \sum X_i^2)$ is sufficient, and so is the one-to-one image $(\bar X, S^2)$. (b) Beta$(\alpha,1)$: $f = \exp\{(\alpha-1)\log x + \log\alpha\}$ on $(0,1)$, so $\sum \log X_i$ is sufficient. (c) When the support depends on θ, use indicators. For the two-parameter exponential $\frac{1}{\sigma}e^{-(x-\mu)/\sigma}\mathbf{1}\{x\ge\mu\}$,

$$\prod f(x_i;\mu,\sigma) = \sigma^{-n}e^{-(\sum x_i - n\mu)/\sigma}\mathbf{1}\{\min x_i \ge \mu\},$$

so $(\sum X_i, X_{(1)})$ is sufficient. For $U[\theta_1,\theta_2]$, $\prod f = (\theta_2-\theta_1)^{-n}\mathbf{1}\{\theta_1 \le x_{(1)}, x_{(n)} \le \theta_2\}$, so $(X_{(1)}, X_{(n)})$ is sufficient.

Theorem 8.5

If the MLE $\hat\theta$ exists uniquely, it is a function of any sufficient statistic $S = u(\mathbf{X})$.

Proof. By factorization, $\hat\theta(\mathbf{x}) = \arg\max_\theta k_1(u(\mathbf{x}),\theta)k_2(\mathbf{x}) = \arg\max_\theta k_1(u(\mathbf{x}),\theta)$, which depends on $\mathbf{x}$ only through $u(\mathbf{x})$. ■

If the unique MLE is itself sufficient, then by this theorem it is a function of every sufficient statistic; such a statistic is called **minimal sufficient**. In Examples 8.5–8.7 the MLEs are of this kind.

Theorem 8.6 (Rao–Blackwell)

Let Y be sufficient for θ, $\hat{\eta}$ an estimator of $\eta(\theta)$ with finite variance, and $\hat{\eta}^{RB} = E[\hat{\eta} \mid Y]$. Then $\hat{\eta}^{RB}$ is a statistic, $E_\theta \hat{\eta}^{RB} = E_\theta \hat{\eta}$, and

$$MSE(\hat{\eta}^{RB}, \theta) \le MSE(\hat{\eta}, \theta) \qquad \text{for all } \theta.$$

Proof. Since the conditional law of $\mathbf{X}$ given Y is free of θ, so is $E[\hat{\eta}(\mathbf{X}) \mid Y]$; thus $\hat{\eta}^{RB}$ is a statistic. By the properties of conditional expectation, $E\hat{\eta}^{RB} = E\hat{\eta}$ and

$$\operatorname{Var} \hat{\eta} = E[\operatorname{Var}(\hat{\eta} \mid Y)] + \operatorname{Var}(E[\hat{\eta} \mid Y]) \ge \operatorname{Var} \hat{\eta}^{RB}.$$

Since the bias is the same, $MSE = \operatorname{Var} + \text{bias}^2$ gives the inequality. ■

Example 8.8

For $U[0, \theta]$, $Y = X_{(n)}$ is sufficient (Example 8.7(c) with $\theta_1 = 0$). Improve $\hat{\theta} = 2\bar{X}$, whose MSE is $\theta^2/(3n)$. Given $X_{(n)} = y$, the other $n-1$ order statistics are distributed as order statistics of $n-1$ uniforms on $(0, y)$, so $E(X_{(r)} \mid X_{(n)} = y) = \frac{r}{n} y$, $r \le n-1$. Hence

$$\hat{\theta}^{RB} = \frac{2}{n} E\Big(\sum_{r=1}^{n} X_{(r)} \,\Big|\, X_{(n)} \Big) = \frac{2}{n} \Big(\sum_{r=1}^{n-1} \frac{r}{n} + 1 \Big) X_{(n)} = \frac{n+1}{n} X_{(n)},$$

with $MSE = \int_0^\theta \left(\frac{n+1}{n} x - \theta\right)^2 \frac{n x^{n-1}}{\theta^n} dx = \frac{\theta^2}{n(n+2)} \le \frac{\theta^2}{3n}$.

8.3 Minimum variance unbiased estimation

Rao–Blackwell improves an estimator by conditioning on a sufficient statistic. If the statistic is also complete, the result is the UMVUE.

Definition (Completeness)

A statistic Y is **complete** for $\theta \in \Omega$ if $E_\theta g(Y) = 0$ for all $\theta \in \Omega$ implies $P_\theta(g(Y) = 0) = 1$ for all θ. A statistic that is complete and sufficient is a **complete sufficient statistic**.

Theorem 8.7 (Lehmann–Scheffé)

Let Y be complete sufficient for $\theta \in \Omega$.

(a) If $\hat{\eta}_0$ is any unbiased estimator of $\eta(\theta)$, then $E[\hat{\eta}_0 \mid Y]$ is the UMVUE of $\eta(\theta)$.

(b) If $\delta(Y)$ is unbiased for $\eta(\theta)$, then $\delta(Y)$ is the UMVUE.

Proof. (a) $\hat{\eta}_0^{RB} = E[\hat{\eta}_0 \mid Y]$ is unbiased. Let $\hat{\eta}_1$ be any other unbiased estimator and $\hat{\eta}_1^{RB} = E[\hat{\eta}_1 \mid Y]$. By Rao–Blackwell, $\operatorname{Var} \hat{\eta}_1^{RB} \le \operatorname{Var} \hat{\eta}_1$. Both $\hat{\eta}_0^{RB}$ and $\hat{\eta}_1^{RB}$ are functions of Y with mean $\eta(\theta)$, so $E_\theta[\hat{\eta}_0^{RB} - \hat{\eta}_1^{RB}] = 0$ for all θ, and by completeness $\hat{\eta}_0^{RB} = \hat{\eta}_1^{RB}$ with probability one. Hence $\operatorname{Var} \hat{\eta}_0^{RB} = \operatorname{Var} \hat{\eta}_1^{RB} \le \operatorname{Var} \hat{\eta}_1$. (b) Apply (a) with $\hat{\eta}_0 = \delta(Y)$, for which $E[\delta(Y) \mid Y] = \delta(Y)$. ■

So, to find a UMVUE: find a complete sufficient statistic Y, then either find a function of Y that is unbiased, or condition a simple unbiased estimator on Y.

Example 8.9 (Bernoulli)

$Y = \sum X_i \sim B(n, \theta)$. If $E_\theta g(Y) = \sum_y g(y)\binom{n}{y}\theta^y(1-\theta)^{n-y} = 0$ for all $0 < \theta < 1$, divide by $(1-\theta)^n$ and set $\lambda = \theta/(1-\theta)$: $\sum_y g(y)\binom{n}{y}\lambda^y = 0$ for all $\lambda > 0$. A polynomial vanishing on an interval has zero coefficients, so $g \equiv 0$. Thus Y is complete sufficient, and $\bar{X} = Y/n$ is the UMVUE of θ.

Example 8.10 (Poisson and exponential)

(a) Poisson: $Y = \sum X_i \sim \text{Poisson}(n\theta)$, and $\sum_y g(y)\lambda^y/y! = 0$ for all $\lambda > 0$ forces $g \equiv 0$ (uniqueness of power series). So $\bar{X}$ is the UMVUE of θ. (b) Exponential: $Y = \sum X_i \sim \text{Gamma}(n, \theta)$, and $\int_0^\infty g(y)y^{n-1}e^{-\lambda y}dy = 0$ for all $\lambda > 0$ forces $g(y)y^{n-1} = 0$ (uniqueness of the Laplace transform). So $\bar{X}$ is the UMVUE of θ.

The general result for exponential families also rests on the uniqueness of Laplace transforms; we state it without proof.

Theorem 8.8

Let $f(x;\eta) = \exp\{\sum_{j=1}^k \eta_j T_j(x) - A(\eta) + S(x)\}$ satisfy: (i) the support does not depend on η; (ii) the parameter space N contains an open box $(a_1, b_1) \times \cdots \times (a_k, b_k)$; (iii) $\mathbf{c}'\mathbf{T}(X)$ is not constant for $\mathbf{c} \neq \mathbf{0}$. Then $\left(\sum_i T_1(X_i), \ldots, \sum_i T_k(X_i)\right)$ is complete sufficient.

Example 8.11

(a) For $N(\mu, \sigma^2)$ the natural parameter $(\mu/\sigma^2, -1/(2\sigma^2))$ ranges over $\mathbb{R} \times (-\infty, 0)$, so $(\sum X_i, \sum X_i^2)$, and hence $(\bar{X}, S^2)$, is complete sufficient. Since $E\bar{X} = \mu$ and $ES^2 = \sigma^2$, they are the UMVUEs.
(b) For $N(\mu, \mu^2)$, $\mu > 0$, the natural parameter $(1/\mu, -1/(2\mu^2))$ lies on a curve, which contains no open box, so Theorem 8.8 does not apply. Indeed $(Y_1, Y_2) = (\sum X_i, \sum X_i^2)$ is not complete: $E_\mu Y_2 = 2n\mu^2$ and $E_\mu Y_1^2 = n(n+1)\mu^2$, so $g = \frac{Y_2}{2n} - \frac{Y_1^2}{n(n+1)}$ has mean 0 for all μ without being 0.
(c) For $U[0, \theta]$, $Y = X_{(n)}$ has pdf ny^{n-1}/θ^n on $(0, \theta)$. If $\int_0^\theta g(y)y^{n-1}dy = 0$ for all $\theta > 0$, differentiating gives $g(\theta)\theta^{n-1} = 0$. So Y is complete, and $\frac{n+1}{n}X_{(n)}$ is the UMVUE of θ (Example 8.8).

Example 8.12 (Bernoulli variance)

To estimate $\eta = \theta(1-\theta)$ from $n \geq 2$ trials: $X_1(1 - X_2)$ is unbiased. Conditioning on $Y = \sum X_i = y$,

$$E[X_1(1-X_2) \mid Y = y] = \frac{P(X_1 = 1, X_2 = 0, \sum_{i \geq 3} X_i = y - 1)}{P(Y = y)} = \frac{\binom{n-2}{y-1}}{\binom{n}{y}} = \frac{y(n-y)}{n(n-1)}.$$

With $\hat{\theta} = Y/n$, the UMVUE is $\frac{n}{n-1}\hat{\theta}(1-\hat{\theta})$.

Example 8.13 (A function of a Poisson mean)

Find the UMVUE of $\eta = e^{-2\theta}$ from a Poisson(θ) sample. We need δ with $E_\theta \delta(Y) = e^{-2\theta}$,

$Y \sim \text{Poisson}(n\theta)$:

$$\sum_{y\ge 0}\delta(y)\frac{e^{-n\theta}(n\theta)^y}{y!} = e^{-2\theta} \iff \sum_y \frac{\delta(y)}{y!}\lambda^y = e^{(1-2/n)\lambda} = \sum_y \frac{(1-2/n)^y}{y!}\lambda^y,$$

with $\lambda = n\theta$. Comparing coefficients, $\delta(y) = (1-2/n)^y$. For large n, $(1-2/n)^{n\bar X} \approx e^{-2\bar X}$, the MLE. But for $n = 1$ the UMVUE is $(-1)^{X_1}$, which takes the value -1 for an estimand in $(0,1)$: insisting on unbiasedness can give absurd answers.

Example 8.14 (Reliability for a normal mean)

For $N(\theta,1)$, $n \ge 2$, estimate $\eta = P_\theta(X_1 > a)$. The indicator $\mathbf{1}\{X_1 > a\}$ is unbiased, and $\bar X$ is complete sufficient, so the UMVUE is $P(X_1 > a \mid \bar X)$. Since $X_1 - \bar X \sim N(0, \frac{n-1}{n})$ is independent of $\bar X$ (Theorem 4.5 in Volume I),

$$P(X_1 > a \mid \bar X = y) = P(X_1 - \bar X > a - y) = 1 - \Phi\Big(\sqrt{\tfrac{n}{n-1}}\,(a-y)\Big),$$

so the UMVUE is $1 - \Phi\big(\sqrt{n/(n-1)}\,(a - \bar X)\big)$.

In this example the independence of $X_1 - \bar X$ and $\bar X$ made the conditional expectation easy. A general source of such independence:

Theorem 8.9 (Basu)

If Y is complete sufficient for θ, and the distribution of a statistic Z does not depend on θ (Z is **ancillary**), then Y and Z are independent.

Proof. Fix B. By sufficiency, $P(Z \in B \mid Y)$ does not depend on θ; and $P_\theta(Z \in B) =: p_B$ does not depend on θ by ancillarity. Then $E_\theta[P(Z \in B \mid Y) - p_B] = 0$ for all θ, so by completeness $P(Z \in B \mid Y) = p_B$ with probability one. That is, $P(Z \in B \mid Y) = P(Z \in B)$ for every B. ∎

Example 8.15 (Reliability for an exponential mean)

For $\text{Exp}(\theta)$, estimate $\eta = P_\theta(X_1 > a) = e^{-a/\theta}$. With $Y = \sum X_i$ complete sufficient, the UMVUE is $P(X_1 > a \mid Y)$. Writing $X_i = \theta Z_i$ with $Z_i \overset{\text{i.i.d.}}{\sim} \text{Exp}(1)$, the ratio $X_1/Y = Z_1/\sum Z_i \sim \text{Beta}(1, n-1)$ is ancillary, hence independent of Y by Basu's theorem. Therefore

$$P(X_1 > a \mid Y = y) = P\Big(\frac{X_1}{Y} > \frac{a}{y}\Big) = \int_{a/y}^1 (n-1)(1-z)^{n-2}dz = \Big(1 - \frac{a}{y}\Big)^{n-1}\mathbf{1}\{y > a\},$$

and the UMVUE is $\big(1 - \frac{a}{n\bar X}\big)^{n-1}\mathbf{1}\{n\bar X > a\}$, close to the MLE $e^{-a/\bar X}$ for large n.

8.4 Asymptotic comparison

For consistent estimators the MSE is $\text{Var} + \text{bias}^2$, and typically the squared bias is $O(n^{-2})$ while the variance is of order n^{-1}. So for large n we compare the variances of the limiting distributions.

Definition (Asymptotic relative efficiency)

If $\sqrt{n}(\hat{\theta}_n^i - \theta) \xrightarrow{d} N(0, \sigma_i^2(\theta))$, $i = 1, 2$, the **asymptotic relative efficiency** of $\hat{\theta}^1$ with respect to $\hat{\theta}^2$ is

$$ARE(\{\hat{\theta}_n^1\}, \{\hat{\theta}_n^2\}) = \frac{\sigma_2^2(\theta)}{\sigma_1^2(\theta)}.$$

A value above 1 means $\hat{\theta}^1$ is more precise. Equivalently, to reach the same accuracy, $\hat{\theta}^2$ needs about ARE times as many observations as $\hat{\theta}^1$: if a 95% error bound $1.96\sigma_i/\sqrt{n_i}$ is to equal d, then $n_i \approx (1.96/d)^2\sigma_i^2$, and $n_2/n_1 = \sigma_2^2/\sigma_1^2$.

Example 8.16 (Location families)

Let $f(x - \theta)$ be a location model with f symmetric about 0. From Chapter 5 and Theorem 6.13,

$$\sqrt{n}(\hat{\theta}^{\text{med}} - \theta) \xrightarrow{d} N\Big(0, \frac{1}{4f(0)^2}\Big), \qquad \sqrt{n}(\bar{X} - \theta) \xrightarrow{d} N(0, \sigma_f^2), \qquad \sqrt{n}(\hat{\theta}^{\text{MLE}} - \theta) \xrightarrow{d} N(0, I_f^{-1}),$$

where $\sigma_f^2 = \int x^2 f$ and $I_f = \int (f'/f)^2 f$. For three standard cases:

	$N(\theta, 1)$	$L(\theta, 1)$	$DE(\theta, 1)$
$1/(4f(0)^2)$ (median)	$\pi/2$	4	1
σ_f^2 (mean)	1	$\pi^2/3$	2
I_f^{-1} (MLE)	1	3	1
ARE(mean, median)	$\pi/2$	$12/\pi^2$	1/2
ARE(median, MLE)	$2/\pi$	3/4	1

For logistic data, the MLE beats the mean, which beats the median. The mean is about 1.2 times as efficient as the median, so the median needs about 20% more observations for the same accuracy.

Example 8.17 (Beta$(\alpha, 1)$)

Here $m_1 = \alpha/(\alpha + 1)$, so $\hat{\alpha}^{\text{MME}} = \bar{X}/(1 - \bar{X})$. With $\sigma^2 = \operatorname{Var} X = \frac{\alpha}{\alpha+2} - \frac{\alpha^2}{(\alpha+1)^2}$ and $g(m) = m/(1 - m)$, $\dot{g}(m_1) = (\alpha + 1)^2$, the delta method gives

$$\sqrt{n}(\hat{\alpha}^{\text{MME}} - \alpha) \xrightarrow{d} N\Big(0, \frac{\alpha(\alpha + 1)^2}{\alpha + 2}\Big).$$

The family is exponential with $T = \log x$; the MLE solves $\sum \log X_i + n/\alpha = 0$, so $\hat{\alpha}^{\text{MLE}} = -1/\overline{\log X}$, with $I(\alpha) = 1/\alpha^2$ and $\sqrt{n}(\hat{\alpha}^{\text{MLE}} - \alpha) \xrightarrow{d} N(0, \alpha^2)$. Hence

$$ARE(\{\hat{\alpha}^{\text{MME}}\}, \{\hat{\alpha}^{\text{MLE}}\}) = \frac{\alpha^2(\alpha + 2)}{\alpha(\alpha + 1)^2} = \frac{\alpha(\alpha + 2)}{(\alpha + 1)^2} < 1.$$

In these examples the MLE had the smallest limiting variance. The next inequality explains why.

Theorem 8.10 (Information inequality)

Assume the regularity conditions (R0)–(R5) of Chapter 6. For any estimator $\hat{\eta}_n$ =

$\hat{\eta}_n(X_1,\dots,X_n)$ of a real quantity with finite variance,

$$\operatorname{Var}_\theta(\hat{\eta}_n) \geq \Big(\frac{\partial}{\partial\theta} E_\theta \hat{\eta}_n\Big)' [nI(\theta)]^{-1} \Big(\frac{\partial}{\partial\theta} E_\theta \hat{\eta}_n\Big).$$

Proof. One-dimensional θ first. By the Cauchy–Schwarz inequality for covariances,

$$\operatorname{Var}_\theta(\hat{\eta}_n)\operatorname{Var}_\theta(\dot{l}_n(\theta)) \geq \{\operatorname{Cov}_\theta(\hat{\eta}_n, \dot{l}_n(\theta))\}^2.$$

By Theorem 6.12, $E_\theta \dot{l}_n(\theta) = nE_\theta \partial_\theta \log f(X_1;\theta) = 0$ and $\operatorname{Var}_\theta \dot{l}_n(\theta) = nI(\theta)$. By (R4),

$$\operatorname{Cov}_\theta(\hat{\eta}_n, \dot{l}_n(\theta)) = E_\theta[\hat{\eta}_n \dot{l}_n(\theta)] = \int \hat{\eta}_n(\mathbf{x}) \frac{\partial}{\partial\theta} \prod f(x_i;\theta)\, d\mathbf{x} = \frac{\partial}{\partial\theta} E_\theta \hat{\eta}_n.$$

Substituting gives the inequality. For vector θ, let $\Sigma_{11} = \operatorname{Var} \hat{\eta}_n$, $\Sigma_{12} = \operatorname{Cov}(\hat{\eta}_n, \dot{l}_n(\theta))' = (\partial_\theta E_\theta \hat{\eta}_n)'$, $\Sigma_{22} = \operatorname{Var} \dot{l}_n(\theta) = nI(\theta)$. Then

$$\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} = \operatorname{Var}\left(\hat{\eta}_n - \Sigma_{12}\Sigma_{22}^{-1}\dot{l}_n(\theta)\right) \geq 0.$$

■

Corollary 8.11 (Cramér–Rao bound)

If $\hat{\theta}^{UE}$ is unbiased for $\theta \in \mathbb{R}^k$, then $\mathbf{c}' \operatorname{Var}_\theta(\hat{\theta}^{UE})\mathbf{c} \geq \mathbf{c}' I(\theta)^{-1}\mathbf{c}/n$ for all $\mathbf{c}$; i.e. $\operatorname{Var}_\theta(\hat{\theta}^{UE}) - I(\theta)^{-1}/n$ is nonnegative definite. For $k = 1$, $\operatorname{Var}_\theta(\hat{\theta}^{UE}) \geq 1/\{nI(\theta)\}$.

Proof. Apply Theorem 8.10 to $\hat{\eta} = \mathbf{c}'\hat{\theta}^{UE}$, for which $\partial_\theta E_\theta \hat{\eta} = \mathbf{c}$. ■

Example 8.18

In Beta$(\alpha, 1)$ (Example 8.17), $Y = \sum \log X_i$ is complete sufficient, and $-\alpha \log X_i \overset{\text{i.i.d.}}{\sim} \text{Exp}(1)$, so $-\alpha Y \sim \text{Gamma}(n, 1)$. Hence

$$E\Big[\frac{1}{-Y}\Big] = \frac{\alpha}{n-1}, \qquad \operatorname{Var}\Big[\frac{1}{-Y}\Big] = \frac{\alpha^2}{(n-1)^2(n-2)}.$$

So $\hat{\alpha}^{\text{UMVUE}} = (n-1)/(-Y) = \frac{n-1}{n}\hat{\alpha}^{\text{MLE}}$, with Var $= \alpha^2/(n-2)$. This exceeds the Cramér–Rao bound α^2/n, but the ratio tends to 1, and $\sqrt{n}(\hat{\alpha}^{\text{UMVUE}} - \alpha) \xrightarrow{d} N(0, \alpha^2)$, the same limit as the MLE.

By the corollary, $\operatorname{Var}\{\sqrt{n}(\hat{\theta}^{UE} - \theta)\} \geq 1/I(\theta)$. This suggests that any asymptotically normal estimator has limiting variance at least $1/I(\theta)$, the limiting variance of the MLE (Theorem 6.13). This is **Fisher's conjecture**. It is not true in complete generality, but it holds under additional conditions such as locally uniform convergence. In this sense the MLE is **asymptotically efficient**.

Example 8.19 (Gamma)

For Gamma(α, β), the MME $(\hat{\alpha}, \hat{\beta})$ (Example 6.1(d)) satisfies

$$\sqrt{n}(\hat{\theta}^{\text{MME}} - \theta) \xrightarrow{d} N(\mathbf{0}, \Sigma(\theta)), \qquad \Sigma(\theta) = \begin{pmatrix} 2\alpha(\alpha+1) & -2(\alpha+1)\beta \\ -2(\alpha+1)\beta & (2+3/\alpha)\beta^2 \end{pmatrix},$$

while the MLE (or its one-step version, Example 6.16) has limit covariance $I(\theta)^{-1}$. Thus

$$ARE(\hat{\alpha}^{\text{MME}}, \hat{\alpha}^{\text{MLE}}) = \frac{1}{2(\alpha+1)\{\alpha\dot{\Psi}(\alpha) - 1\}},$$

$$ARE(\hat{\beta}^{\mathrm{MME}}, \hat{\beta}^{\mathrm{MLE}}) = \frac{\alpha\dot{\Psi}(\alpha)}{(2\alpha+3)\{\alpha\dot{\Psi}(\alpha)-1\}}.$$

For $\alpha = 1$ these are about 0.39 and 0.51; they increase towards 1 as $\alpha \to \infty$ but are always below 1. Generally, the MME loses efficiency relative to the MLE.